

Supervised Machine Learning

Numerical Problem Bank: Worked Examples & Student Exercises | Unit Core Focus

1. LINEAR REGRESSION

KEY EQUATIONS & PARAMETERS

Model: $\hat{y} = w_0 + w_1x$ | **Slope (w_1):** $w_1 = \sum((x_i - \bar{x})(y_i - \bar{y})) / \sum((x_i - \bar{x})^2) = \text{Cov}(x, y) / \text{Var}(x)$

Intercept (w_0): $w_0 = \bar{y} - w_1\bar{x}$ | **Means:** $\bar{x} = (\sum x_i)/n$, $\bar{y} = (\sum y_i)/n$

WORKED EXAMPLE 1.1 – ORDINARY LEAST SQUARES FIT

Problem Statement

Given 3 observations of Study Hours (x) and Exam Scores (y): (2, 4), (4, 7), (6, 10). Compute the parameters w_1 and w_0 , and predict score for $x = 5$ hours.

i	x_i	y_i	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
1	2	4	-2	-3	6	4
2	4	7	0	0	0	0
3	6	10	2	3	6	4
Sum	12	21	0	0	12	8

- Calculate Means:** $\bar{x} = 12 / 3 = 4$, $\bar{y} = 21 / 3 = 7$
- Calculate Slope w_1 :** $w_1 = 12 / 8 = 1.5$
- Calculate Intercept w_0 :** $w_0 = \bar{y} - w_1\bar{x} = 7 - (1.5 \times 4) = 7 - 6 = 1.0$
- Final Regression Equation:** $\hat{y} = 1.0 + 1.5x$
- Prediction for $x = 5$:** $\hat{y} = 1.0 + 1.5(5) = 8.5$

STUDENT PRACTICE 1.2

Exercise 1

A sales dataset contains three records for Advertising Spend (\$k) vs Sales (\$k): (1, 3), (3, 5), (5, 10).

Task: (a) Calculate sample means $\bar{x}$ and $\bar{y}$. (b) Find the OLS regression line $\hat{y} = w_0 + w_1x$. (c) Predict sales for an advertising spend of \$4k.

2. POLYNOMIAL REGRESSION

KEY EQUATIONS & CONCEPT

Degree-d Polynomial Model: $\hat{y} = w_0 + w_1x + w_2x^2 + \dots + w_dx^d$

Models non-linear feature relationships while keeping linear parameters with respect to weight vector w .

WORKED EXAMPLE 2.1 – NON-LINEAR FEATURE EVALUATION

Problem Statement

A quadratic regression model trained on non-linear physical data yields the hypothesis: $\hat{y} = 3.0 - 1.5x + 2.0x^2$. Compute predicted targets for inputs $x \in \{-2, 0, 3\}$.

- For $x = -2$:** $\hat{y} = 3.0 - 1.5(-2) + 2.0(-2)^2 = 3.0 + 3.0 + 2.0(4) = 14.0$
- For $x = 0$:** $\hat{y} = 3.0 - 1.5(0) + 2.0(0)^2 = 3.0$
- For $x = 3$:** $\hat{y} = 3.0 - 1.5(3) + 2.0(3)^2 = 3.0 - 4.5 + 2.0(9) = 16.5$

3. DECISION TREE REGRESSION

KEY EQUATIONS

Leaf Prediction Rule: $\hat{y} = (1/N) \sum_{i \in \text{Leaf}} y_i$ (Mean of target values in leaf region)

Node Variance: $\text{Var}(S) = (1/N) \sum_{i=1}^N (y_i - \bar{y})^2$ | **Variance Reduction:** $\Delta \text{Var} = \text{Var}(\text{Parent}) - \sum (N_{\text{child}}/N) \text{Var}(\text{Child})$

WORKED EXAMPLE 3.1 – LEAF PREDICTION & VARIANCE

Problem Statement

During decision tree construction, a candidate leaf node receives 5 continuous targets: $Y = [4, 6, 7, 9, 14]$.

Tasks: (a) Determine the prediction value $\hat{y}$ for this leaf node. (b) Calculate sample variance in this leaf.

1. **Calculate Mean Prediction ($\hat{y}$):**

$$\hat{y} = (4 + 6 + 7 + 9 + 14) / 5 = 40 / 5 = 8.0$$

2. **Calculate Squared Deviations from Mean:**

$$(4-8)^2 = 16, (6-8)^2 = 4, (7-8)^2 = 1, (9-8)^2 = 1, (14-8)^2 = 36$$

3. **Sum of Squared Errors (SSE):** $16 + 4 + 1 + 1 + 36 = 58$

4. **Sample Node Variance:** $\text{Var}(Y) = 58 / 5 = 11.6$

STUDENT PRACTICE 3.2

Exercise 3

A parent node with target set $Y = [2, 4, 6, 8]$ is split into two children: Left $Y_L = [2, 4]$ and Right $Y_R = [6, 8]$.

Task: Compute variance of the parent node, variance of each child node, and total Variance Reduction ΔVar .

4. LOGISTIC REGRESSION

KEY EQUATIONS

Logits (z): $z = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_n x_n$

Sigmoid Function: $P(y=1|x) = \sigma(z) = 1 / (1 + e^{-z})$ | **Decision Rule:** Class 1 if $\sigma(z) \geq 0.5$ else Class 0

WORKED EXAMPLE 4.1 – BINARY CLASSIFICATION PROBABILITY

Problem Statement

A binary logistic regression classifier has parameters $w_0 = -1.5$, $w_1 = 0.6$, and $w_2 = 1.2$. Calculate log-odds z , predicted probability p , and class label for input vector $X = [x_1 = 2, x_2 = 1]$.

1. **Compute Logit (z):**

$$z = -1.5 + (0.6 \times 2) + (1.2 \times 1) = -1.5 + 1.2 + 1.2 = 0.90$$

2. **Compute Probability $p = \sigma(0.90)$:**

$$p = 1 / (1 + e^{-0.90}) = 1 / (1 + 0.4066) = 1 / 1.4066 \approx 0.7109 \text{ (71.09\%)}$$

3. **Apply Threshold (0.5):**

Since $p = 0.7109 \geq 0.5$, the output is classified as **Class 1**.

STUDENT PRACTICE 4.2

Exercise 4

Given weights $w_0 = 1.0$, $w_1 = -0.5$ for feature $x = 4$. (Take $e^1 \approx 2.718$).

Task: Calculate z , posterior probability $P(y=1|x)$, and decide the final class (Threshold = 0.5).

5. K-NEAREST NEIGHBORS (KNN)

KEY DISTANCE METRIC

Euclidean Distance: $d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Classification assigns majority vote among K samples with smallest distance d .

WORKED EXAMPLE 5.1 — DISTANCE & MAJORITY VOTE

Problem Statement

Given 2D training data with labels: **A(1, 1, Red), B(2, 3, Red), C(4, 2, Blue), D(3, 4, Blue)**. Classify query point **T(2, 2)** for $K = 1$ and $K = 3$.

Point	x_1	x_2	Class	Distance Formula $d(T, \text{Point})$	Computed Distance
A	1	1	Red	$\sqrt{((1-2)^2 + (1-2)^2)} = \sqrt{1+1}$	$\sqrt{2} \approx 1.414$
B	2	3	Red	$\sqrt{((2-2)^2 + (3-2)^2)} = \sqrt{0+1}$	1.000
C	4	2	Blue	$\sqrt{((4-2)^2 + (2-2)^2)} = \sqrt{4+0}$	2.000
D	3	4	Blue	$\sqrt{((3-2)^2 + (4-2)^2)} = \sqrt{1+4}$	$\sqrt{5} \approx 2.236$

- Ranked Distances:** Point B (1.000) < Point A (1.414) < Point C (2.000) < Point D (2.236)
- For $K = 1$:** Nearest neighbor is B (Red) → Prediction: **Red**
- For $K = 3$:** 3 Nearest neighbors are {B(Red), A(Red), C(Blue)} → Votes: Red=2, Blue=1 → Prediction: **Red**

STUDENT PRACTICE 5.2

Exercise 5

Given training points: **P1(0, 0, Class 0), P2(2, 0, Class 0), P3(0, 3, Class 1), P4(3, 3, Class 1)**.

Task: Classify query point **Q(1, 1)** using K-Nearest Neighbors for $K = 3$.

6. DECISION TREE CLASSIFIER & RANDOM FOREST

KEY IMPURITY METRICS

Gini Impurity: $Gini = 1 - \sum_{i=1}^C p_i^2$ | **Entropy:** $H(S) = -\sum_{i=1}^C p_i \log_2(p_i)$

Random Forest: Ensembles M trees trained on bootstrap samples; predicts via Majority Vote $\hat{y} = \text{Mode}(\hat{y}_1, \dots, \hat{y}_M)$.

WORKED EXAMPLE 6.1 — IMPURITY CALCULATIONS

Problem Statement

A candidate node in a binary classifier contains 10 samples: **6 Class Positive (+), 4 Class Negative (-)**. Compute both Gini Impurity and Entropy for this node.

- Probabilities:** $p_+ = 6/10 = 0.6$, $p_- = 4/10 = 0.4$
- Gini Impurity:**
 $Gini = 1 - (0.6^2 + 0.4^2) = 1 - (0.36 + 0.16) = 1 - 0.52 = 0.48$
- Entropy:**
 $H(S) = -[0.6 \log_2(0.6) + 0.4 \log_2(0.4)]$
 $\log_2(0.6) \approx -0.737$, $\log_2(0.4) \approx -1.322$
 $H(S) = -[0.6(-0.737) + 0.4(-1.322)] = -[-0.4422 - 0.5288] = 0.971 \text{ bits}$

STUDENT PRACTICE 6.2

Exercise 6

7. NAÏVE BAYES CLASSIFIER

KEY THEOREM

Bayes' Theorem with Conditional Independence Assumption:

$$P(Y|x_1, \dots, x_n) \propto P(Y) \prod_{i=1}^n P(x_i|Y) \quad | \text{Normalized by dividing by } P(X).$$

WORKED EXAMPLE 7.1 – TEXT CLASSIFICATION INFERENCE

Problem Statement

Dataset for Spam detection with priors $P(\text{Spam}) = 0.4$, $P(\text{Ham}) = 0.6$. Given word likelihoods: $P(\text{"Offer"}|\text{Spam}) = 0.8$, $P(\text{"Offer"}|\text{Ham}) = 0.1$. Determine if a message containing "Offer" is Spam or Ham.

1. **Unnormalized Posteriors:** $P^*(\text{Spam}) = 0.8 \times 0.4 = 0.32$, $P^*(\text{Ham}) = 0.1 \times 0.6 = 0.06$
2. **Normalized Posteriors:** $P(\text{Spam}|\text{Offer}) = 0.32 / 0.38 = 0.8421$ (84.21%), $P(\text{Ham}|\text{Offer}) = 15.79\%$
3. **Decision:** Classify as **Spam**.

STUDENT PRACTICE 7.2

Exercise 7

Priors: $P(\text{Buy}) = 0.3$, $P(\text{NoBuy}) = 0.7$. Likelihoods for Age=Young: $P(\text{Young}|\text{Buy}) = 0.7$, $P(\text{Young}|\text{NoBuy}) = 0.2$.

Task: Compute normalized posterior probability $P(\text{Buy}|\text{Young})$.

8. CONCEPTS OF BIAS, VARIANCE, UNDERFITTING & OVERFITTING

CORE CONCEPTUAL FRAMEWORK

Expected Total Error: $\text{Expected Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$

High Bias: Simple model, underfitting (High train & test error). **High Variance:** Complex model, overfitting (Low train error, High test error).

WORKED EXAMPLE 8.1 – SCENARIO IDENTIFICATION

Problem Statement

Analyze model behaviors on a dataset and diagnose model state:

- **Model A (Degree 1):** Train Error = 35%, Test Error = 36% → **Underfitting (High Bias)**
- **Model B (Degree 15):** Train Error = 1%, Test Error = 28% → **Overfitting (High Variance)**
- **Model C (Degree 3):** Train Error = 5%, Test Error = 6% → **Optimal Model Fit**

STUDENT PRACTICE 8.2

Exercise 8

A Decision Tree trained without depth limits achieves 100% training accuracy but only 62% testing accuracy on unseen data.

Task: Identify whether this model suffers from high bias or high variance, and explain how constraining tree depth affects both error components.